Conjecture 00000001149: an impossible norm for a Gaussian coefficient

lizaixi01, with OpenAI Codex assistance

22 October 2026

1 The necessary condition already fails

The conjecture asks for a p-adically convergent series whose mth term is a q-binomial coefficient with absolute value $p^{-m/2}$, for coprime minimal-model parameters p, q. Take $p = 2$ and $q = 5$. These satisfy $2 \leq p < q$ and $\gcd(p, q) = 1$. Regardless of which fusion coefficient is to be represented, the term with $m = 1$ would have to satisfy

$$|a_1|_2 = 2^{-1/2}.$$

This is impossible for any $a_1 \in \mathbf{Q}_2$, hence in particular for a Gaussian coefficient at the integer parameter $q = 5$. The argument applies whether the series is numbered from zero or one, since both conventions include the term indexed by one.

Indeed, every nonzero $x \in \mathbf{Q}_2$ has an integer valuation $v_2(x)$ and $|x|_2 = 2^{-v_2(x)}$. If this were $2^{-1/2}$, injectivity of the real exponential with base 2 would give $2v_2(x) = 1$, a contradiction. Zero also cannot have that norm. Thus no sequence meeting the asserted termwise condition exists; questions about its convergence and its sum do not arise.

2 The coefficients are actual rational expressions

We use the standard Gaussian coefficient, not a free variable assigned a convenient type. In the notation of NIST DLMF, equation 17.2.27,

$$(q;q)_n = \prod_{j=0}^{n-1}(1 - q^{j+1}), \qquad \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_k (q;q)_{n-k}} \quad (0 \leq k \leq n).$$

At $q = 5$ all factors in the denominator are nonzero rational numbers, so every displayed coefficient is rational. We set the coefficient to zero for $k > n$, which cannot help achieve the required nonzero norm either. Integrality of Gaussian coefficients is a stronger classical fact, but is unnecessary for this obstruction.

The Lean source defines these finite products and quotients over $\mathbf{Q}$, proves their denominators nonzero, and maps the result into the actual field `Padic 2`. It also considers the canonical image in `PadicAlgCl 2`; the extended norm agrees with the $\mathbf{Q}_2$ norm. Therefore passing to the algebraic closure does not alter the norm of any of these coefficients.

3 Formal statement and scope

`no_half_integral_norm` proves the obstruction for every element of $\mathbf{Q}_2$. The Gaussian-coefficient definitions then give `gaussianPadic_ne_half_norm` and its algebraic-closure version. Finally, `no_requested_gaussian_series` quantifies over arbitrary natural-number upper and lower index sequences $n(m), k(m)$ and rules out the required norm law for all $m \geq 1$. The theorem `counterexample` combines this with the explicit legal parameter check for $(2, 5)$.

This disproves a necessary clause of the question as printed. Here q is the integer minimal-model parameter appearing in the same statement, and the coefficients are the standard, unscaled Gaussian coefficients. In conformal-field-theory literature the letter q may instead denote a separate nome. That alternative parameter convention is not specified in this question and is not asserted to be covered by this argument. It neither defines nor evaluates the Virasoro fusion coefficients, and makes no claim about a modified norm law, a different meaning of a generalized q-binomial coefficient, or arbitrary algebraic numbers outside the rational-coefficient image. The obstruction does not assert that $2^{-1/2}$ is absent from the norm-value set of the entire algebraic closure.

4 Verification and provenance

The accompanying project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

The source-bound build, transitive axiom audit and official fresh kernel replay are recorded in the package evidence. No `sorry`, `native_decide`, or extra axiom is used. The source statement is preserved at its upstream commit in `SOURCE.md`. OpenAI Codex assisted the proof and report; a separate supervising Codex agent reviewed the formal chain. This is AI review, not human certification. Organizer acceptance is pending.